

SILIN RAJ GUPTA

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Education

University of Waterloo

Bachelor of Computer Science, Honours, Co-op

Waterloo, ON

Sep 2024 – Apr 2029 (Expected)

Skills

Languages: Python, C/C++, JavaScript/TypeScript, Ruby, Swift, SQL, HTML/CSS

AI / ML: LLMs (OpenAI, Anthropic, Gemini, Grok), RAG & vector search (Pinecone), prompt engineering, agent evals, computer vision, speech (STT/TTS, realtime voice), LangChain

Web & Mobile: React, Next.js, React Native (Expo), Node.js, FastAPI, Rails, GraphQL (Apollo), Tailwind CSS

Tools & Infrastructure: Git, Docker, Linux, Bash, CI/CD, AWS, GCP, PostgreSQL/Supabase, n8n, Figma

Professional Experience

Applied Machine Learning Engineer

May 2026 – Aug 2026

Shopify (Search Product & Tooling)

Toronto, ON

- Contributed to open source 🐙 on **Tangle** 🌐, Shopify's ML experimentation platform: authored an **agent-skills bundle** (8-step tuning loop, 7 subagents) that drives coding agents through end-to-end ML experiments.
- Authored a **library of 17 reference skills** codifying reusable patterns for agent-authored pipelines, including secret handling that teaches Tangent to resolve credentials from secret stores instead of inlining them.
- Built an evaluation suite that regression-tests the **8 agentic skills** guiding coding agents (Claude Code, Pi) to author, submit, and review ML pipelines on Tangle.
- Designed **75 planted-defect fixtures** scoring whether the agent catches seeded faults, measuring skill improvements vs. regressions per SKILL.md edit and auto-proposing LLM-generated skill fixes for failing cases.
- Parallelized the suite's sequential test-case runs into concurrent execution, cutting full eval time from **4 hours to 28 minutes**.

Full Stack AI Engineer

Jan 2026 – Apr 2026

E3 Group

San Francisco, CA

- Re-architected a production freight-invoice-to-**EDI 210** pipeline (Python, Gemini vision OCR), replacing a single monolithic prompt with versioned prompts and broker-specific post-processing modules validated against EDI ground truth, lifting field-extraction accuracy from **70% to 95%**.
- Developed an automated **email-to-EDI ingestion pipeline** handling **~600 invoices/day** from Gmail, classifying and merging attachments and resolving carriers via parallel Gemini extraction, removing a third-party FTP/OCR dependency.
- Engineered the reliability layer that let EDIs ship without manual review: a build-time **verification gate** flagging fields absent from the source document and **cross-drop deduplication** that ended a week-long duplicate-billing incident, replacing next-day spreadsheet QA.
- Built a shared **React component library** and agent skills that migrate a dashboard's full UI in one prompt, then shipped **multiple client-tailored dashboards** for invoice review and annotation.
- Led **weekly 1:1 client calls** to triage issues and prioritize fixes, scaling from **4 to 20+ broker/carrier feeds** through production cutover.

iOS App Developer

Oct 2025 – Dec 2025

Merazine

San Francisco, CA

- Built a full-stack AI companion app (React Native, GraphQL, Supabase) with **real-time voice** over OpenAI's Realtime API and dual-tier memory, plus ESP32-S3 firmware for the companion hardware (BLE WiFi provisioning, I2S mic capture, bidirectional PCM audio streaming).

Software Engineer

May 2025 – Aug 2025

PolicyAdvisor

Toronto, ON

- Built a Next.js + FastAPI proposal-automation platform on the Google Gemini API, cutting processing from **5–6 hrs to 2–5 mins** at 99%+ accuracy across 10+ carriers and saving **~\$100,000/year**.
- Spearheaded **viQ** 🌐, a visitor-insurance platform integrating 10+ carrier APIs with real-time pricing and caching, leading a cross-functional team to **launch in 2 weeks**.
- Orchestrated a **24/7 AI voicebot** (n8n, ElevenLabs, Twilio, Google Calendar) handling client calls and auto-scheduling callbacks, reducing missed leads by **40%**.

Projects

🐙 **SpeakSummarize - Record. Recall** 🌐 🍏 | [1K+ Downloads, \$1K MRR] Swift, Whisper-1, GPT-4o-mini, Pinecone, mem0

- Built and shipped an AI speech-to-notes iOS app (Whisper-1 transcription, GPT-4o-mini summaries, multi-speaker diarization), reaching **1,000+ downloads** and **\$1K MRR** within days of launch.
- Grew via Reddit and mem0-based preference learning, now extending to Android and watchOS.

🐙 **Muse Sketch Studio** 🌐 | [Replicate AI Hackathon Winner] React.js, TypeScript, Node.js, Replicate API

- Built an **end-to-end AI fashion design pipeline** (sketch → colors → model → 3D → runway video) from text prompts via Google Nano-Banana and Kling v2.1.